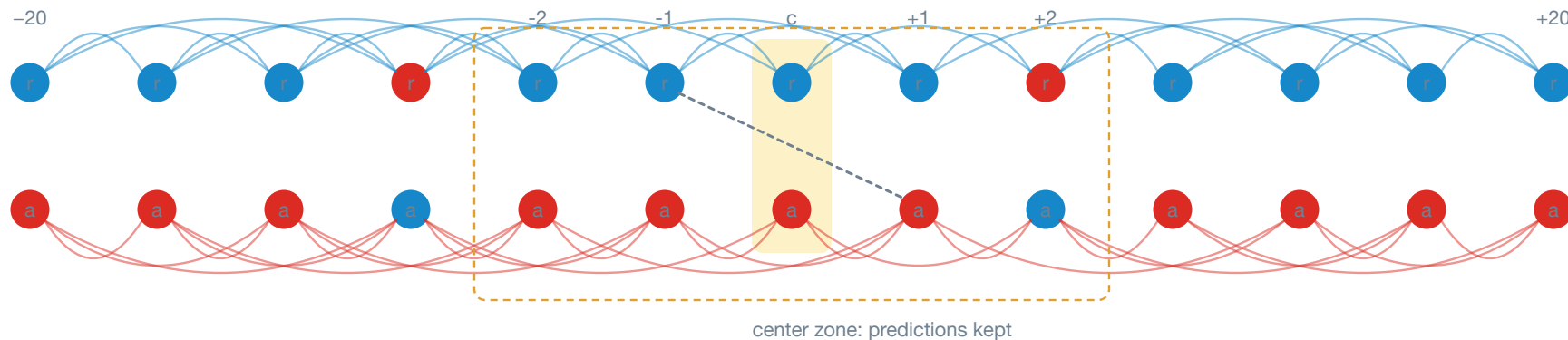


a A window is opened around every phased variant with $PE \geq 0.8$



- up to 20 variants on each side (≤ 41 variants, ≤ 82 allele nodes; windows above 256 nodes are skipped)
- nodes: one per allele; SNV, indel, SV and 5mC records are merged and sorted by position
- edges: DOT edges with both endpoints in the window, symmetrized, plus one self-loop per node (mean of incident edge features)
- predictions of the two allele nodes are averaged; a variant covered by several windows receives the mean

b Bridge vertex feature



c Center-zone mask

$r = |\text{pos} - \text{pos}_c| / \text{max offset in window}$
kept if $r \leq \min(1, 10 / \text{round}(20 \cdot r_{\text{max}}))$
so that predictions are read only where both sides of the variant are visible to the network.